

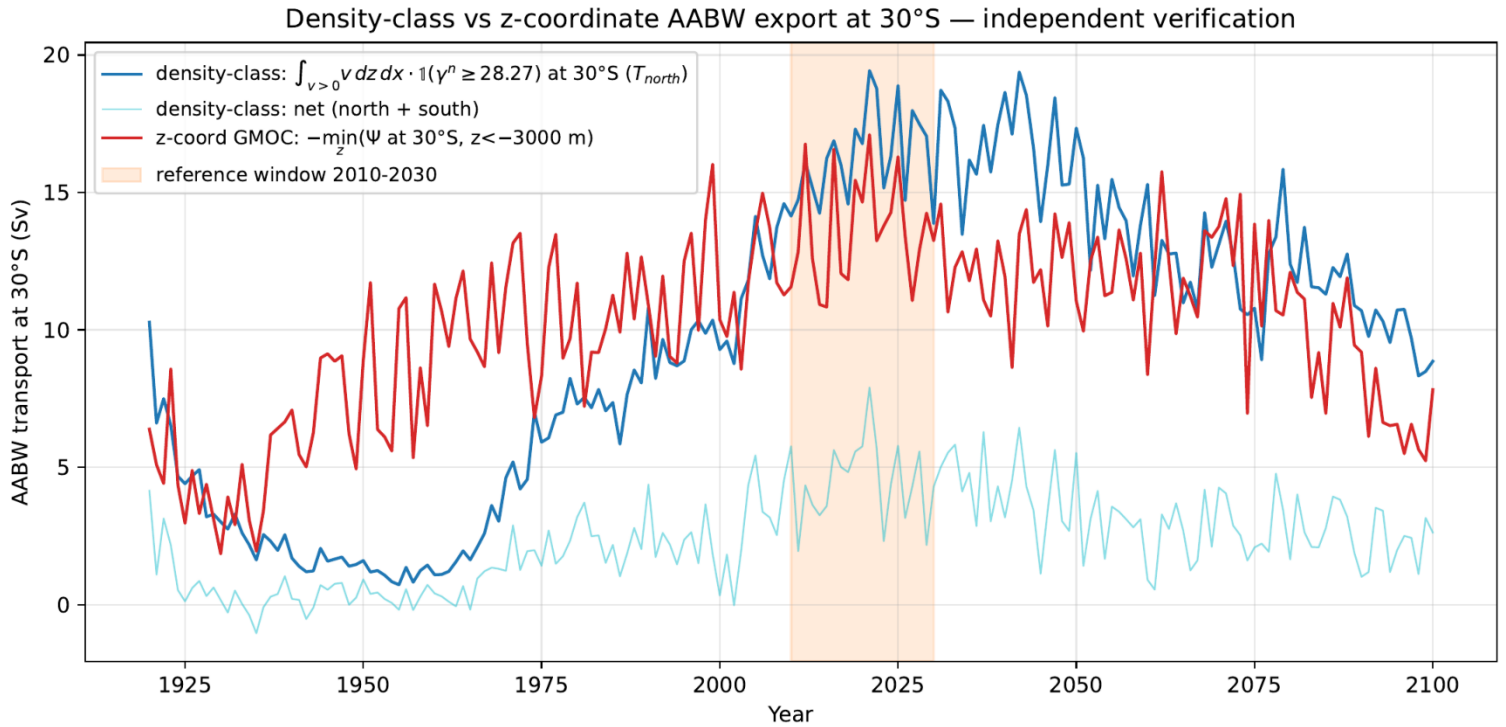
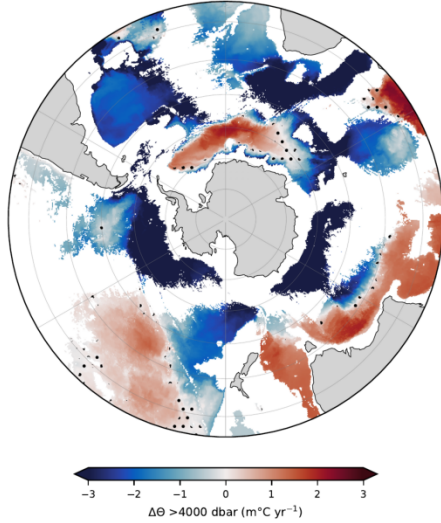


Figure 1. Independent verification of the AABW transport at 30°S using density-class versus z-coordinate diagnostics.

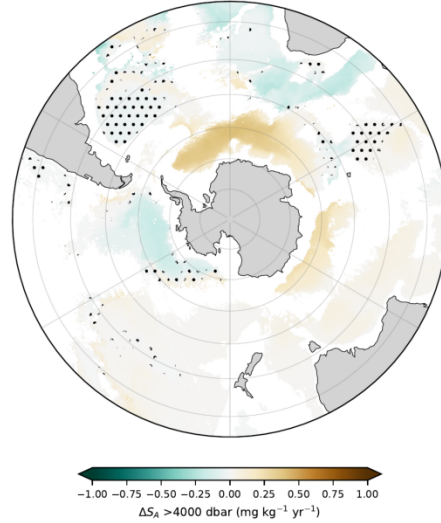
FESOM2 hres simulation, 1920–2100 (April–October mean). The dark-blue curve shows the northward density-class AABW export, computed as a strict element-wise cross-section integral across 30°S, $T_{north} = \int_{\{v>0\}} v \, dz \, dx \cdot \mathbb{1}(\gamma^n \geq 28.27)$, evaluated on the native FESOM2 triangular mesh (5,144 elements crossed by 30°S; chord length $\times$ layer thickness $\times$ meridional velocity, restricted to neutral density $\gamma^n \geq 28.27 \text{ kg m}^{-3}$). The light-blue curve is the density-class net transport (north + south, retaining the southward return flow). The red curve is the AABW intensity from the z-coordinate global MOC, defined as $-\min_z \Psi$ at 30°S below 3000 m. The orange band marks the 2010–2030 reference window where both estimates show their joint peak. Both independent diagnostics reproduce the non-monotonic 21st-century evolution: a rise from $\sim 5 \text{ Sv}$ (1920s) to a peak of 12–15 Sv near 2025, followed by a weakening to 6–8 Sv by 2100. The 2–4 Sv offset between methods is consistent with the z-coordinate streamfunction extremum being a more conservative integral measure than the density-bounded export.

FESOM2 hres — Fig.4 hist (1920–2014) vs proj (2015–2100)
 cols: deep Θ (>4000 dbar), deep S_A (>4000 dbar), AABW thk ($\Theta_{CT}=0^\circ\text{C}$)
 (hatching: MK $p > 0.05$)

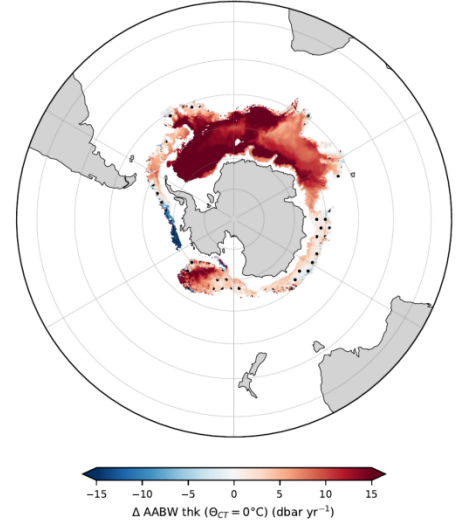
a 1920–2014



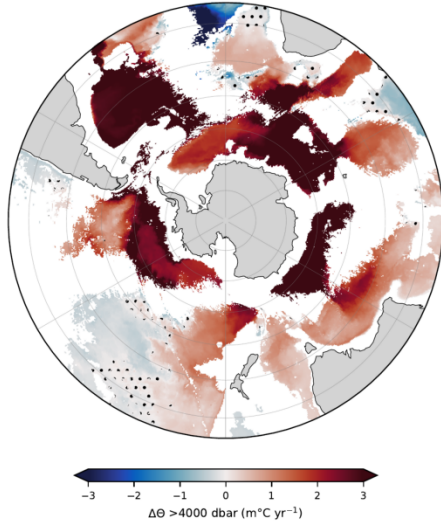
b 1920–2014



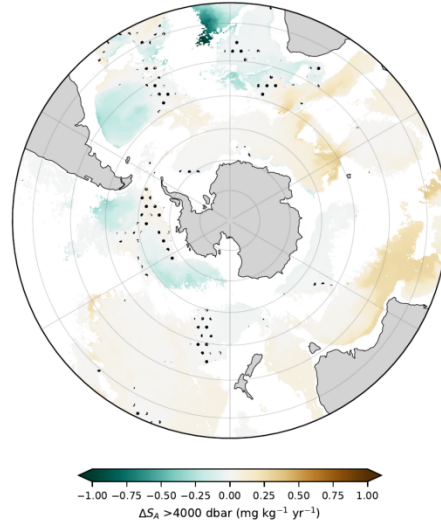
c 1920–2014



d 2015–2100



e 2015–2100



f 2015–2100

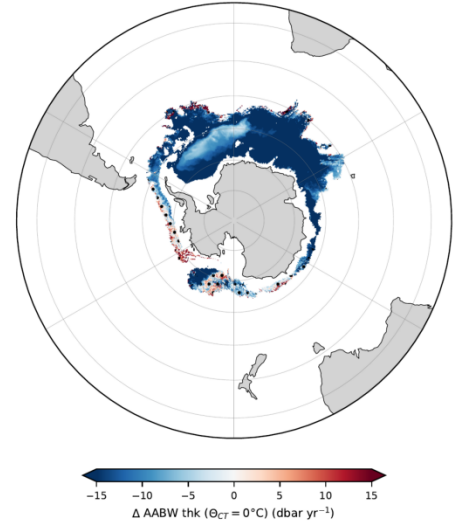


Figure 2. Sign reversal of deep-Southern-Ocean trends between historical and projection periods, FESOM2 hres simulation.

Rows: historical 1920–2014 (a–c) versus projection 2015–2100 (d–f). Columns: (left) volume-weighted potential temperature trend below 4000 dbar ($\text{m}^\circ\text{C yr}^{-1}$); (centre) volume-weighted absolute salinity S_A trend below 4000 dbar ($\text{mg kg}^{-1} \text{yr}^{-1}$); (right) AABW-layer thickness trend defined as the pressure span below the $\Theta_{CT} = 0^\circ\text{C}$ isotherm (dbar yr^{-1} , after Purkey & Johnson 2010 / Rintoul 2026), shown for nodes with bathymetry deeper than 3000 m. Stippling marks Mann–Kendall non-significance ($p > 0.05$). The historical period shows mostly cooling, freshening-then-salinification, and AABW thickening (53% of points; median $+1.6 \text{ dbar yr}^{-1}$). In the projection, signs reverse: deep waters warm coherently across all sectors (median $+0.40 \text{ m}^\circ\text{C yr}^{-1}$), S_A trends weaken, and AABW shrinks across 96% of the domain (median $-6.84 \text{ dbar yr}^{-1}$), evidencing accelerated 21st-century deep-ocean warming as the driver of AABW contraction.

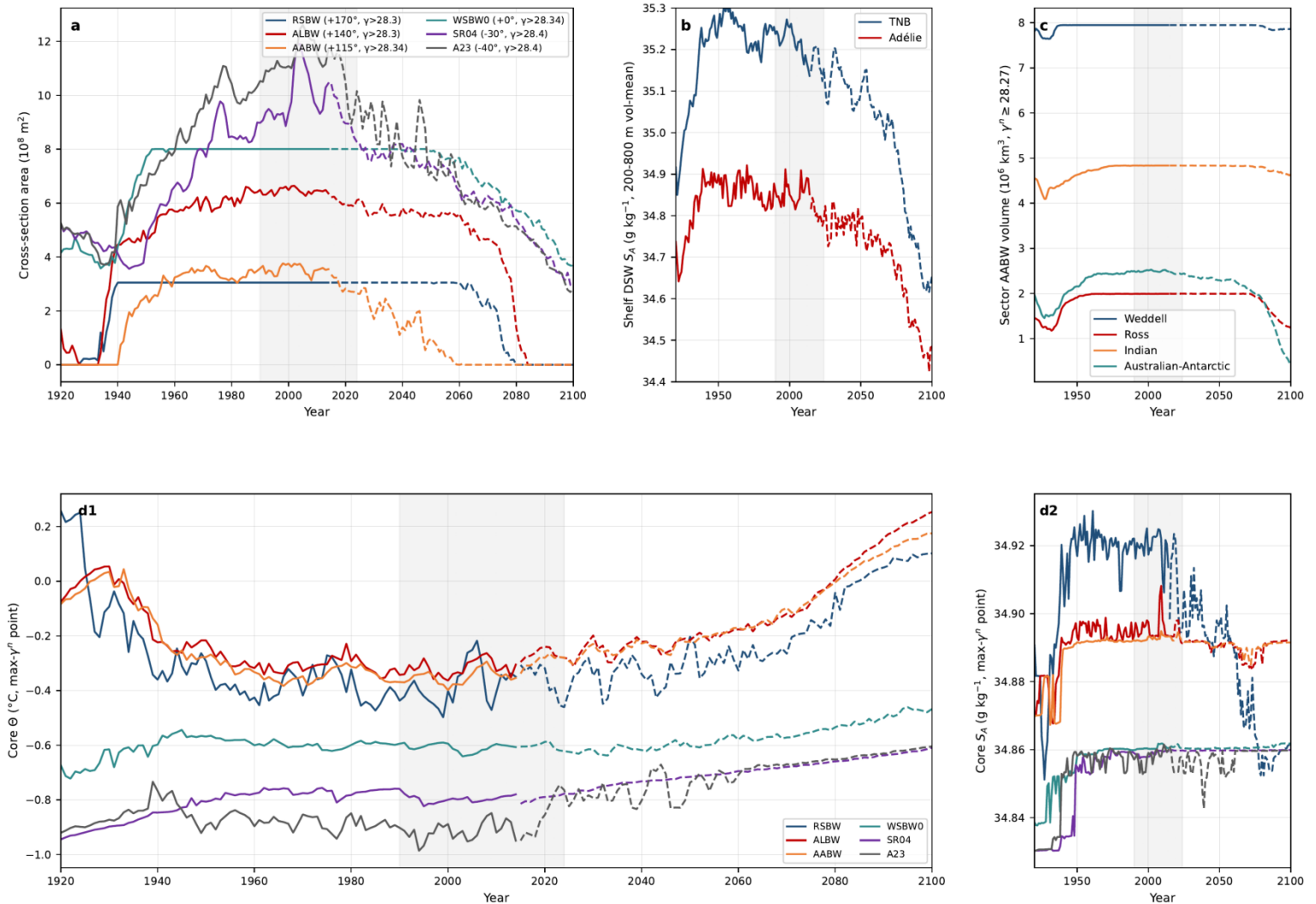


Figure 3. Reproduction of Rintoul et al. (2026) Fig. 5 with FESOM2 hres, 1920-2100 (April-October mean).

(a) Density-class cross-section areas (10^8 m^2) of six bottom-water transects: Ross Sea Bottom Water (RSBW, 170°E , $\gamma^\theta > 28.3$), Adélie Land Bottom Water (ALBW, 140°E), AABW (115°E), Weddell Sea Bottom Water (WSBW0, 0°), and the SR04/A23 sections in the Weddell Gyre (with progressively higher density thresholds). (b) Shelf Dense-Shelf-Water S_A (g kg^{-1} , volume-mean 200–800 m) for Terra Nova Bay (TNB) and Adélie Land — the two principal Antarctic source-water polynyas. (c) Sector-integrated AABW volume (10^6 km^3 , $\gamma^\theta \geq 28.27$) for the Weddell, Ross, Indian, and Australian-Antarctic sectors. (d1, d2) Core Θ and S_A at the maximum- γ^θ point of each transect. Solid: historical (1920–2014); dashed: SSP projection (2015–2100); shaded grey: 1990–2020 reference period. All sectors show coherent late-21st-century declines in cross-section area, shelf DSW salinity, and AABW volume, accompanied by core warming of $0.3\text{--}1.0 \text{ }^\circ\text{C}$; Amundsen–Bellingshausen-adjacent transects (RSBW, ALBW) collapse first, while the Weddell sector resists until ~ 2070 .

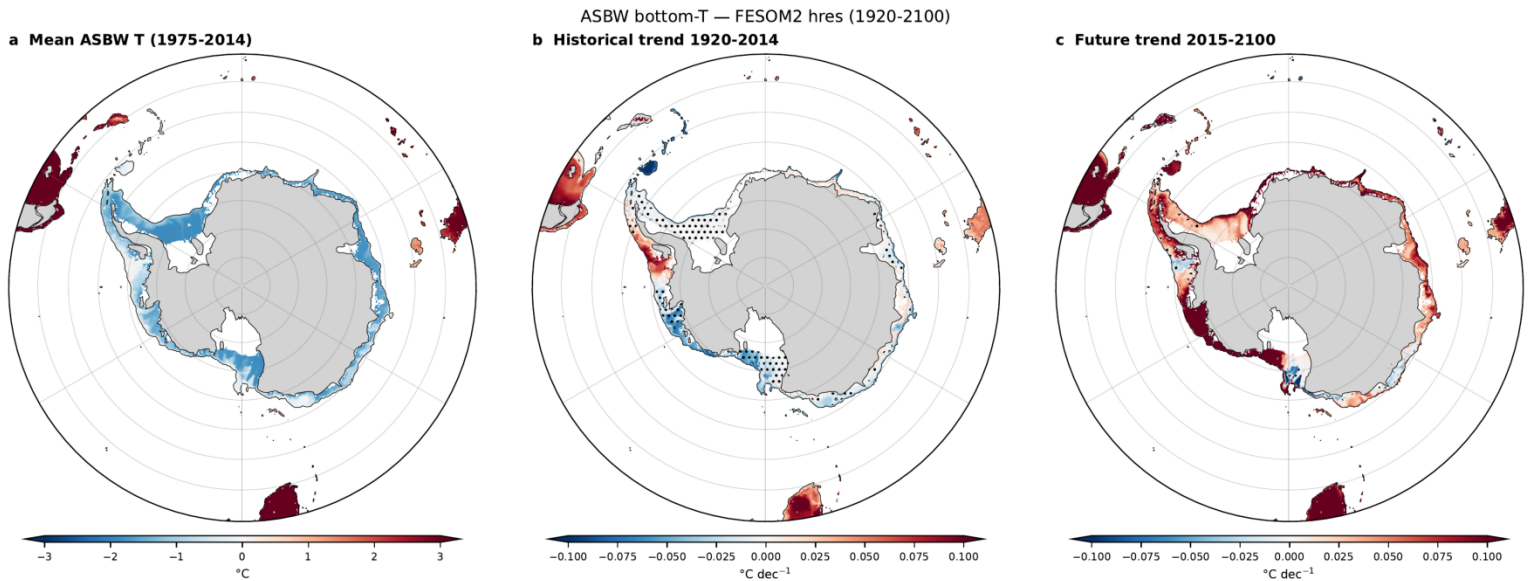


Figure 4. Antarctic Shelf Bottom Water (ASBW) bottom-temperature climatology and trends, FESOM2 hres simulation (after Purich et al. 2021 Fig. 2).

(a) Mean bottom-T over the 1975–2014 baseline (°C); (b) historical linear trend 1920–2014 (°C dec⁻¹); (c) projected linear trend 2015–2100 (°C dec⁻¹). Stippling in (b) and (c) marks Mann–Kendall non-significance ($p > 0.05$). The ASBW mask follows Purich et al. (2021): bottom layer on the Antarctic continental shelf, restricted to $T < 0$ °C in the climatology. Historical trends are spatially heterogeneous with weak cooling-warming dipoles around the Antarctic Peninsula and modest signals elsewhere. The projection shows a circumpolar near-uniform warming, strongest in the Amundsen–Bellingshausen and East Antarctic sectors (>0.1 °C dec⁻¹), consistent with intrusion of warmer Circumpolar Deep Water onto the shelves and the late-21st-century shutdown of cold dense-water formation.

FESOM2 hres — regional ASBW temperature anomaly time series
(baseline: 1950-1979 mean)

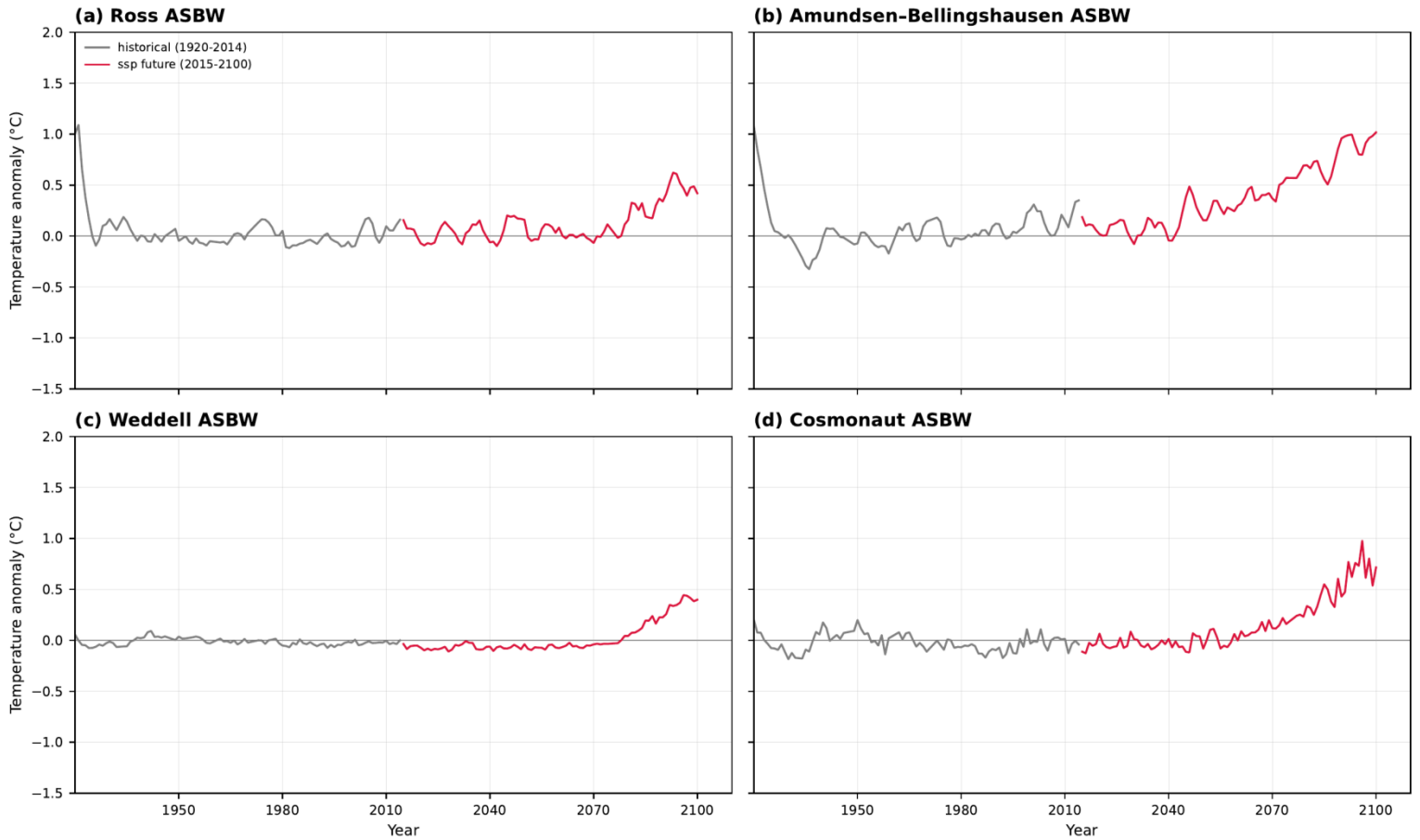


Figure 5. Regional ASBW temperature-anomaly time series, FESOM2 hres simulation, 1920-2100 (after Purich et al. 2021 Fig. 3).

Annual ASBW bottom-temperature anomalies (°C) averaged over four Antarctic shelf regions: (a) Ross, (b) Amundsen-Bellingshausen, (c) Weddell, (d) Cosmonaut. Anomalies are taken relative to the 1950-1979 climatology of each region. Grey: historical run (1920-2014); red: SSP projection (2015-2100). All four regions remain near baseline through the 20th century. Warming emerges earliest and is largest in the Amundsen-Bellingshausen sector ($\sim +1.0$ °C by 2100), followed by Cosmonaut ($\sim +0.8$ °C). The Ross and Weddell shelves stay close to their historical climatology until ~ 2060 - 2070 , then warm by $\sim +0.4$ - 0.5 °C by 2100, consistent with the regional onset sequence of dense-water-formation shutdown described in Fig. 3.